

Department of Computer Science and Engineering
Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar
First Sessional Exam
Subject: Machine Learning II
Paper Code: BCE-P624

Time: 1 Hr

MM: 20

Note: Answer any two questions from Section A (each question carries 5 marks)

Answer any one question from Section B (each question carries 10 marks)

Q.No.	Section A (Questions of 5 marks each)
Q1.	What is Unsupervised Learning and how does it differ from Supervised Learning?
Q2.	Given 10 features, each taking 5 possible values, and 3 classes. how many parameters are required without and with Naïve Bayes assumption?
Q3.	What is classification? Explain how input is interpreted in classification and describe how categorical features are handled.
	Section B (Questions of 10 marks each)
Q1.	Consider the following 3-dimensional dataset with data points classified into three classes: Blue, Green and Red Instance(ID): {A,B,C,D,E,F,G,H,I} Feature1(X1): {1,1,4,5,6,,2,3,4,2} Feature1(X3): {1,2,5,2,1,2,3,2,3} Feature1(X2): {1,3,2,3,1,2,3,1,1} Class(Y): {Blue,Blue,Green,Green,Red,Red,Red,Green,Red} Classify a new data point with features(X1=3,X2=2,X=3) using K-Nearest Neighbors with K=3 using the following distance metrics: 1)Euclidean distance 2)Manhattan 3)Chebyshev 4)Cosine Similarity
Q2.	Consider the following 2D Dataset with six data point. Data point: {1,2,3,4,5,6} Feature(X1): {1,1.5,5,8,1,9} FeatureX2: {2,1.8,8,,8,0.6,11} Let's choose K=2 and apply k-means algorithm with initially chosen centroid as data point C1=1 and C2=3.

Section A – Short Answer (Attempt any two questions 5 Marks each)		CO	BL																																																																		
Q-1	How Does a Decision Tree Work?	CO2	B2																																																																		
Q-2	(a) Define the Sigmoid Function. • Calculate its output for $z=0$, $z=-2$ and $z=-2$. (b) Given input vector is $z=[2,1,0]$ Find 1. Compute the softmax probabilities for each element. 2. Verify that the sum of probabilities equals 1.	CO2	B3																																																																		
Q-3	What is Clustering and how is it different from classification?	CO4	B2																																																																		
Section B – Long Answer (Attempt any one questions 10 Marks each)		CO	BL																																																																		
Q-1	Given the following dataset: <table><thead><tr><th>Day</th><th>Weather</th><th>Temperature</th><th>Humidity</th><th>Wind</th><th>Play?</th></tr></thead><tbody><tr><td>1</td><td>Sunny</td><td>Hot</td><td>High</td><td>Weak</td><td>No</td></tr><tr><td>2</td><td>Cloudy</td><td>Hot</td><td>High</td><td>Weak</td><td>Yes</td></tr><tr><td>3</td><td>Sunny</td><td>Mild</td><td>Normal</td><td>Strong</td><td>Yes</td></tr><tr><td>4</td><td>Cloudy</td><td>Mild</td><td>High</td><td>Strong</td><td>Yes</td></tr><tr><td>5</td><td>Rainy</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr><tr><td>6</td><td>Rainy</td><td>Cool</td><td>Normal</td><td>Strong</td><td>No</td></tr><tr><td>7</td><td>Rainy</td><td>Mild</td><td>High</td><td>Weak</td><td>Yes</td></tr><tr><td>8</td><td>Sunny</td><td>Hot</td><td>High</td><td>Strong</td><td>No</td></tr><tr><td>9</td><td>Cloudy</td><td>Hot</td><td>Normal</td><td>Weak</td><td>Yes</td></tr><tr><td>10</td><td>Rainy</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr></tbody></table> Tasks: 1. Calculate Entropy of the target variable PlayTennis. 2. Compute Information Gain for the attribute Outlook. 3. Determine the root node of the decision tree. 4. Show the first level of the decision tree after splitting by the best attribute.	Day	Weather	Temperature	Humidity	Wind	Play?	1	Sunny	Hot	High	Weak	No	2	Cloudy	Hot	High	Weak	Yes	3	Sunny	Mild	Normal	Strong	Yes	4	Cloudy	Mild	High	Strong	Yes	5	Rainy	Mild	High	Strong	No	6	Rainy	Cool	Normal	Strong	No	7	Rainy	Mild	High	Weak	Yes	8	Sunny	Hot	High	Strong	No	9	Cloudy	Hot	Normal	Weak	Yes	10	Rainy	Mild	High	Strong	No	CO4	B4
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Q-2	Write a Python program to build and evaluate regression models. Tasks: 1. Load the California Housing dataset using sklearn.datasets. 2. Perform preprocessing: - Handle missing values (if any) - Scale features using StandardScaler 3. Split the dataset into training and testing sets. 4. Train the following models: a) Linear Regression b) Random Forest Regressor 5. Predict outputs on test data. 6. Evaluate both models using: - RMSE (Root Mean Squared Error) - R^2 Score 7. Compare both models and state which performs better.	CO5	B4																																																																		